

Module 5: Action — Testing Your Predictions

Learning Objectives

1. Understand that doing math — solving problems, running calculations, conducting experiments — is how you **test your predictions** against reality.
2. Learn the basics of **experimental design**: how to set up fair tests that actually answer your questions.
3. Discover that getting the "wrong answer" is **valuable data** that tells you where to improve your model.

Introduction

A weather forecaster predicts a 70% chance of rain. A sports analyst predicts your team will win by 10 points. Your math model predicts that studying an extra hour will raise your test score by 5 points. These are all predictions — but how do you know if they're any good?

You test them. You check whether it actually rained. You watch the game. You study the extra hour and see what happens to your grade. Without testing, a prediction is just a guess with fancy math behind it.

In the world of Math Detectives, **action** means putting your mathematical predictions to the test. It's the moment where your model meets reality, and you find out whether you're right, wrong, or somewhere in between.

Key Concepts

1. Every Calculation Is a Test

When you solve a math problem, you're testing a prediction. You predict that your method will produce the correct answer. When you check your work (or when the teacher grades it), you find out whether your prediction was right.

This is why checking your work matters so much. It's not just about getting the right grade — it's about **calibrating your model**. If you consistently make the same type of mistake (say, forgetting to carry in multiplication), that's a systematic prediction error. Once you identify it, you can fix it. But if you never check, you never find the error, and your model stays broken.

Smart math students build in checks: Does the answer make sense? Is it in the right ballpark? Can I verify it a different way? Each check is a mini-experiment testing whether your calculation matched reality.

2. Designing Fair Tests

Not all tests are created equal. If you want to know whether a new study method works better than your old one, you need a **fair test**.

A fair test changes only **one variable** at a time. If you switch to a new study method AND start sleeping more AND get a new tutor all at the same time, and your grades go up, you don't know which change caused the improvement. You've changed too many variables.

Basic experimental design: - **Variable you're testing** (the one you change on purpose): Try studying with flashcards instead of re-reading notes - **Control** (what stays the same): Same subject, same amount of study time, same sleep schedule - **What you measure** (the outcome): Test scores

This is how scientists work, and it's how you can work too. Want to know if listening to music helps you focus on homework? Set up a fair test: do homework with music for a week, without music for a week, keep everything else the same, and compare results.

3. Wrong Answers Are Data

In school, wrong answers usually feel bad. But in the Math Detective framework, wrong answers are **information**.

When your prediction is wrong, the SIZE and DIRECTION of the error tell you something useful. If you predicted 50 and the answer was 500, you probably made a fundamental error in your approach. If you predicted 50 and the answer was 52, your approach was solid and just needed a small adjustment.

Scientists call this **residual analysis** — studying the gap between predictions and reality to improve the model. It's one of the most powerful tools in all of mathematics.

The same principle applies to everyday predictions. If the weather forecast said 0% chance of rain and it poured, that's a massive prediction error that means the weather model needs serious revision. If it said 70% chance and it didn't rain, that's actually fine — 70% means it won't rain 30% of the time. Understanding the difference between a bad prediction and normal uncertainty is a crucial mathematical skill.

Try It!

Prediction Scorecard. For one week, make one testable prediction each day. They can be about anything measurable: "I predict I'll finish my homework in 30 minutes," "I predict at least 3 people will be wearing red tomorrow," "I predict the temperature will be above 60 degrees." Record your prediction AND the actual result. At the end of the week, calculate your prediction accuracy. Were your predictions too high? Too low? What patterns do you see in your errors?

Summary

Mathematical action means **testing your predictions** through calculation, experimentation, and observation. Good tests require **fair experimental design** — changing one variable at a time. When predictions are wrong, the errors are **valuable data** that help you improve your models. The goal isn't to always be right; it's to get better at predicting over time. Next: **Learning** — how to update your mathematical models based on what your tests reveal.